

exercise-bank

Typst Package

Exercise Management & Banking System

A comprehensive solution for creating, organizing, and filtering exercises

Version 0.5.0

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1. Introduction

`exercise-bank` is a Typst package for creating and managing exercises with solutions, metadata, filtering, and exercise banks. Perfect for teachers, textbook authors, and educational content creators.

1.1. Features

- Exercises with inline or deferred solutions
- Teacher corrections with flexible display modes
- Draft mode for work-in-progress documents
- Multiple display control options (what, where, which content)
- Metadata support (topic, level, author, competencies)
- Exercise banks: define once, display anywhere
- Powerful filtering by any criteria
- Automatic numbering with reset options
- Customizable labels (localization support)

1.2. Installation

Import the package in your Typst document:

```
#import "@preview/exercise-bank:0.5.0": exo, exo-setup
```

1.3. Quick Start

Code

```
#exo(  
  exercise: [Solve the equation  $2x + 5 = 13$ .]  
)
```

Preview

Exercise 1 Solve the equation $2x + 5 = 13$.

2. Basic Usage

2.1. Simple Exercise

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.]  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

2.2. Exercise with Solution

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.],  
  solution: [$3 + 4 times 2 = 3 + 8 = 11$],  
)
```

Preview

Exercise 1 Calculate $3 + 4 \times 2$.

Solution 1 $3 + 4 \times 2 = 3 + 8 = 11$

2.3. Multiple Exercises

Exercises are automatically numbered:

Code

```
#exo(exercise: [Simplify $x^2 + 2x + 1$.])  
#exo(exercise: [Factor $x^2 - 4$.])  
#exo(exercise: [Solve $2x - 6 = 0$.])
```

Preview

Exercise 1 Simplify $x^2 + 2x + 1$.

Exercise 2 Factor $x^2 - 4$.

Exercise 3 Solve $2x - 6 = 0$.

3. Display Control

The package uses three parameters to control display behavior.

3.1. display - What to Display

Controls what content is displayed:

- "both" (default) - Show exercises and solutions/corrections
- "ex" - Show only exercises
- "sol" - Show only solutions/corrections

Code

```
#exo-setup(display: "ex") // Exercises only

#exo(
  exercise: [Solve  $x + 3 = 7$ .],
  solution: [ $x = 4$ ], // Hidden
)
```

Preview

Exercise 1 Solve $x + 3 = 7$.

3.2. corr-display - Which Content to Show

Controls whether to show solutions or corrections:

- "solution" (default) - Show solution content
- "correction" - Show correction content
- "mixed" - Default to solution, but use correction for exercises with `show-corr: true`

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Simplify  $2x + 3x$ .],
  correction: [
     $2x + 3x = 5x$ 
    (Combine like terms)
  ],
)
```

Preview

Exercise 1 Simplify $2x + 3x$.

Correction 1 $2x + 3x = 5x$ (Combine like terms)

3.3. corr-loc - Where to Display

Controls where solutions/corrections appear:

- "after" (default) - Immediately after each exercise
- "pagebreak" - With a page break between
- "end-section" - Collected at section end
- "end-chapter" - Collected at chapter end

Code

```
#exo-setup(corr-loc: "end-section")

#exo(exercise: [Exercise 1], solution: [Answer 1])
#exo(exercise: [Exercise 2], solution: [Answer 2])

#exo-print-solutions() // Print collected solutions
```

Preview

Exercise 1 Exercise 1

Exercise 2 Exercise 2

Solutions

Solution 1 Answer 1

Solution 2 Answer 2

4. Corrections (Teacher Version)

Corrections are detailed solutions for teachers, including pedagogical notes and teaching tips.

4.1. Exercise with Correction

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Solve  $x^2 = 9$ .],
  correction: [
    *Teacher notes:*
     $x = \pm 3$ 

    Common mistake: forgetting the negative root.
  ],
)
```

Preview

Exercise 1 Solve $x^2 = 9$.

4.2. Solutions Only (Answer Key)

Show only solutions for student answer keys:

Code

```
#exo-setup(display: "sol")

#exo(
  exercise: [This is hidden],
  solution: [Only this solution is shown],
)
```

Preview

Solution 1 Only this solution is shown

4.3. The sol-in-corr Flag

When `corr-display: "correction"`, both correction AND solution are shown by default. Use `sol-in-corr: true` on an exercise to indicate the solution is already embedded in the correction (avoiding duplication):

Code

```
#exo-setup(corr-display: "correction")

// Without sol-in-corr: both correction and solution shown
#exo(
  exercise: [Problem A],
  correction: [Teacher notes only],
  solution: [$x = 5$],
)

// With sol-in-corr: only correction shown
#exo(
  exercise: [Problem B],
  correction: [
    *Solution:* $x = 3$

    _Teaching tip: Watch for sign errors._
  ],
  solution: [$x = 3$],
  sol-in-corr: true, // Solution already in correction
)
```

Preview

Exercise 1 Problem A

Exercise 2 Problem B

4.4. Mixed Display Mode

Use `corr-display: "mixed"` to default to solutions while showing corrections for specific exercises:

Code

```
#exo-setup(corr-display: "mixed")

#exo(
  exercise: [Simple problem],
  solution: [Quick answer],
  correction: [Detailed explanation],
)

#exo(
  exercise: [Complex problem],
  solution: [Answer],
  correction: [Step-by-step solution],
  show-corr: true, // Shows correction
)
```

Preview

Exercise 1 Simple problem

Exercise 2 Complex problem

5. Draft Mode

Show placeholders for incomplete exercises during document preparation.

5.1. With Draft Mode ON

Code

```
#exo-setup(  
  draft-mode: true,  
  solution-placeholder: [_[To be written]_] ,  
)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - shows placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

5.2. With Draft Mode OFF (Default)

Empty solutions show minimal space without placeholders:

Code

```
#exo-setup(draft-mode: false)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - no placeholder  
)
```

Preview

Exercise 1 Solve $x + 5 = 12$

6. Exercise Banks

Define exercises once, display them anywhere. Perfect for creating reusable exercise collections.

6.1. Defining Bank Exercises

Use `exo-define` to register exercises without displaying them:

```
#exo-define(  
  id: "quad-1",  
  exercise: [Solve  $x^2 - 5x + 6 = 0$ .],  
  topic: "quadratics",  
  level: "1M",  
  solution: [ $x = 2$  or  $x = 3$ ],  
)  
  
#exo-define(  
  id: "geom-1",  
  exercise: [Find the area of a circle with radius 5.],  
  topic: "geometry",  
  level: "1M",  
  solution: [ $A = 25\pi$ ],  
)
```

6.2. Displaying Bank Exercises

Use `exo-show` to display a specific exercise:

Code

```
#exo-show("quad-1")
```

Preview

Exercise 1 Solve $x^2 - 5x + 6 = 0$.

6.3. Selecting Multiple Exercises

Use `exo-select` with filters:

```
// All quadratics exercises  
#exo-select(topic: "quadratics")  
  
// Level 1M exercises only  
#exo-select(level: "1M")  
  
// Multiple topics  
#exo-select(topics: ("quadratics", "geometry"))  
  
// Limit count  
#exo-select(topic: "algebra", max: 5)  
  
// Custom filter  
#exo-select(where: ex => ex.metadata.level == "hard")
```

7. Metadata and Filtering

7.1. Adding Metadata

Tag exercises for organization and filtering:

```
#exo(  
  exercise: [Solve  $x + 1 = 5$ .],  
  topic: "algebra",  
  level: "easy",  
  authors: ("Prof. Smith",),  
)
```

7.2. Filtering Exercises

Display only exercises matching criteria:

```
#exo-filter(topic: "algebra")  
#exo-filter(level: "easy")  
#exo-filter(topic: "algebra", level: "hard")
```

8. Competency Tags

Tag exercises with competencies and display them visually.

Code

```
#exo-setup(show-competencies: true)

#exo-define(
  id: "comp-ex",
  exercise: [Solve and explain your reasoning.],
  competencies: ("C1.1", "C2.3", "C4.1"),
  solution: [Detailed solution here],
)

#exo-show("comp-ex")
```

Preview

Exercise 1 Solve and explain your reasoning.

C1.1 C2.3 C4.1

8.1. Filter by Competency

```
#exo-select(competency: "C1.1")
#exo-select(competencies: ("C1.1", "C2.3"))
```

9. Configuration

9.1. Global Setup

Use `exo-setup` to configure defaults:

```
#exo-setup(  
  // Display control  
  display: "both",           // "ex", "sol", "both"  
  corr-display: "solution",  // "solution", "correction", "mixed"  
  corr-loc: "after",         // "after", "pagebreak", "end-section", "end-chapter"  
  // Labels  
  exercise-label: "Exercise",  
  solution-label: "Solution",  
  correction-label: "Correction",  
  // Counter behavior  
  counter-reset: "section",  // "section", "chapter", "global"  
  // Display options  
  show-id: false,  
  show-competencies: false,  
  // Draft mode  
  draft-mode: false,  
  // Badge styling  
  badge-style: "box",  
  badge-color: black,  
)
```

9.2. Badge Styles

Change the visual style of exercise badges:

```
// Circled number (no label text)  
#exo-setup(badge-style: "circled")  
  
// Filled circle with white number  
#exo-setup(badge-style: "filled-circle", badge-color: rgb("#2563eb"))  
  
// Pill-shaped badge  
#exo-setup(badge-style: "pill")  
  
// Arrow tag  
#exo-setup(badge-style: "tag", badge-color: rgb("#1e40af"))
```

Available styles: "box" (default), "circled", "filled-circle", "pill", "tag", "border-accent", "underline", "rounded-box", "header-card"

9.3. Localization

Change labels for different languages:

Code

```
// French  
#exo-setup(  
  exercise-label: "Exercice",  
  solution-label: "Solution",  
  correction-label: "Corrigé",  
)
```

Preview

French: Exercice, Solution, Corrigé
German: Aufgabe, Lösung
Spanish: Ejercicio, Solución

9.4. Counter Reset Options

Control when numbering resets:

```
#exo-setup(counter-reset: "section") // Reset each section
#exo-setup(counter-reset: "chapter") // Reset each chapter
#exo-setup(counter-reset: "global")  // Never reset
```

Use hooks to trigger resets:

```
= New Section
#exo-section-start() // Resets if counter-reset: "section"

= New Chapter
#exo-chapter-start() // Resets if counter-reset: "chapter"
```

9.5. Advanced Exercises

Mark exercises as advanced to display a visual cue before the label. The default symbol is `*`.

Code

```
#exo(
  exercise: [A challenging problem.],
  advanced: true,
)
```

Preview

*** Exercise 1** A challenging problem.

Customize the symbol:

Code

```
#exo-setup(advanced-symbol: sym.dagger)
#exo(
  exercise: [Advanced with dagger.],
  advanced: true,
)
```

Preview

† Exercise 1 Advanced with dagger.

Disable with `advanced-symbol: none`.

10. Visual Styles

Set the badge style with `exo-setup(badge-style: "...")`.

10.1. Box (Default)

Code

```
#exo-setup(badge-style: "box")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

10.2. Circled

Code

```
#exo-setup(badge-style: "circled")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

10.3. Filled Circle

Code

```
#exo-setup(  
  badge-style: "filled-circle",  
  badge-color: rgb("#2563eb"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

1 Solve $x + 3 = 7$

10.4. Pill

Code

```
#exo-setup(badge-style: "pill")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

10.5. Tag

Code

```
#exo-setup(  
  badge-style: "tag",  
  badge-color: rgb("#1e40af"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve $x + 3 = 7$

10.6. Border Accent

Code

```
#exo-setup(  
  badge-style: "border-accent",  
  badge-color: rgb("#3b82f6"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1
Solve $x + 3 = 7$

10.7. Underline

Code

```
#exo-setup(badge-style: "underline")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve $x + 3 = 7$

10.8. Rounded Box

Code

```
#exo-setup(badge-style: "rounded-box")
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

10.9. Header Card

Code

```
#exo-setup(badge-style: "header-card", badge-color: rgb("#3b82f6"))
#exo(exercise: [Solve $x + 3 = 7$])
```

Preview

Exercise 1

Solve $x + 3 = 7$

11. Parameter Reference

11.1. `exo` Function

Parameter	Type	Default	Description
<code>exercise</code>	content	none	Exercise content (required)
<code>solution</code>	content	none	Solution content
<code>correction</code>	content	none	Correction for teachers
<code>id</code>	string	auto	Unique exercise ID
<code>sol-in-corr</code>	bool	false	If true, solution is already in correction (don't show both)
<code>show-corr</code>	bool	false	If true, show correction in “mixed” mode
<code>topic</code>	string	none	Topic metadata
<code>level</code>	string	none	Difficulty level
<code>authors</code>	array	()	Author names

11.2. `exo-define` Function

Same as `exo`, plus:

Parameter	Type	Default	Description
<code>competencies</code>	array	()	Competency tags
<code>points</code>	number	none	Points (for exam mode)

11.3. `exo-select` Function

Parameter	Type	Default	Description
<code>topic</code>	string	none	Filter by topic
<code>level</code>	string	none	Filter by level
<code>author</code>	string	none	Filter by author
<code>competency</code>	string	none	Filter by competency
<code>topics</code>	array	none	Filter by any topic
<code>levels</code>	array	none	Filter by any level
<code>competencies</code>	array	none	Filter by any competency
<code>where</code>	function	none	Custom filter function
<code>renumber</code>	bool	true	Renumber sequentially
<code>max</code>	int	none	Maximum exercises

11.4. `exo-setup` Function

Display Control:

Parameter	Type	Default	Description
<code>display</code>	string	“both”	“ex”, “sol”, “both”
<code>corr-display</code>	string	“solution”	“solution”, “correction”, “mixed”

corr-loc	string	“after”	“after”, “pagebreak”, “end-section”, “end-chapter”
exercise-label	string	“Exercise”	Label for exercises
solution-label	string	“Solution”	Label for solutions
correction-label	string	“Correction”	Label for corrections
counter-reset	string	“section”	“section”, “chapter”, “global”
show-id	bool	false	Show exercise IDs
show-competencies	bool	false	Show competency tags
draft-mode	bool	false	Show placeholders for empty content
solution-placeholder	content	<i>To be completed</i>	Placeholder text
correction-placeholder	content	<i>To be completed</i>	Placeholder text

Visual Styling:

Parameter	Type	Default	Description
badge-style	string	“box”	Style: “box”, “circled”, “filled-circle”, “pill”, “tag”, “border-accent”, “underline”, “rounded-box”, “header-card”
badge-color	color	black	Color for exercise badges
solution-color	color	green	Color for solution badges
correction-color	color	green	Color for correction badges
label-font-size	length	12pt	Font size for badge labels
margin-position	length/auto	auto	Width reserved for badge column
label-extra	length	1cm	Extra space for labels in margin

Spacing:

Parameter	Type	Default	Description
exercise-above	length	0.8em	Space above exercise boxes
exercise-below	length	0.8em	Space below exercise boxes
solution-above	length	0.8em	Space above solution boxes
solution-below	length	0.8em	Space below solution boxes
correction-above	length	0.8em	Space above correction boxes
correction-below	length	0.8em	Space below correction boxes
advanced-symbol	content/none	“*”	Symbol before label for advanced exercises

12. Utility Functions

Function	Description
<code>exo-reset-counter()</code>	Reset exercise numbering to 0
<code>exo-clear-registry()</code>	Clear all registered exercises
<code>exo-section-start()</code>	Call at section start (triggers reset if configured)
<code>exo-chapter-start()</code>	Call at chapter start (triggers reset if configured)
<code>exo-print-solutions()</code>	Print collected solutions (for end-section/chapter modes)
<code>exo-count(topic: ..)</code>	Count exercises matching criteria
<code>exo-show("id")</code>	Display exercise by ID
<code>exo-show-many("a", "b")</code>	Display multiple exercises by ID
<code>exo-filter(topic: ..)</code>	Filter and display matching exercises

13. Exam Integration

These functions support building exam documents from an exercise bank, typically used alongside a layout package such as `texam`.

13.1. exam-setup

Configure the exam correction box appearance and behaviour.

Parameter	Type	Default	Description
<code>show-solutions</code>	bool	false	Show <code>exam-solution-box</code> content
<code>solution-label</code>	string	"Correction"	Label shown in bold at the top of the solution box
<code>solution-fill</code>	color	green tint	Background fill of the solution box
<code>solution-stroke</code>	stroke	green stroke	Border stroke of the solution box
<code>solution-radius</code>	length	3pt	Corner radius of the solution box
<code>solution-inset</code>	length	8pt	Inner padding of the solution box
<code>solution-label-color</code>	color	dark green	Color of the label text
<code>question-spacing</code>	length	1em	Vertical space between questions

13.2. exam-solution-box

A green-highlighted answer block. Only rendered when `exam-setup(show-solutions: true)`.

```
#exam-solution-box[
  $(x - 2)(x - 3)$
]
```

Place it after an exercise. The same source file produces both the student version (box hidden) and the correction (box shown) by flipping `show-solutions`.

13.3. exam-score-table

Builds a score grid from registered exercise IDs, reading point values from the registry.

```
#exam-score-table("ex-1", "ex-2", "ex-3")
```

Parameter	Type	Default	Description
<code>..exercise-ids</code>	strings		Positional — IDs of exercises to include
<code>question-label</code>	string	"Question"	Header for the question column
<code>points-label</code>	string	"Points"	Header for the points row
<code>grade-label</code>	string	"Note"	Header for the grade row
<code>total-label</code>	string	"Total"	Label for the total column